

DESIGN & BUILD TUTORIAL

FM Radio Lab

Build Your Own

FM Station

A tunable FM transmitter and its matching super-regenerative receiver — two transistors on one board, four on the other, and a hand-wound coil they share. The transmitter's own junction capacitance performs the modulation; the receiver IS the transmitter, strangled 50,000 times a second. Every stage is proven by simulation before you solder it.

Designed with tscircuit · React for electronics

37 math checks · 21 simulation proofs · 75 parts · zero DRC errors · October 2026

FM 88-108 MHZ · COLPITTS · SUPER-REGENERATIVE DETECTOR · SLOPE DETECTION

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1 • What you are building

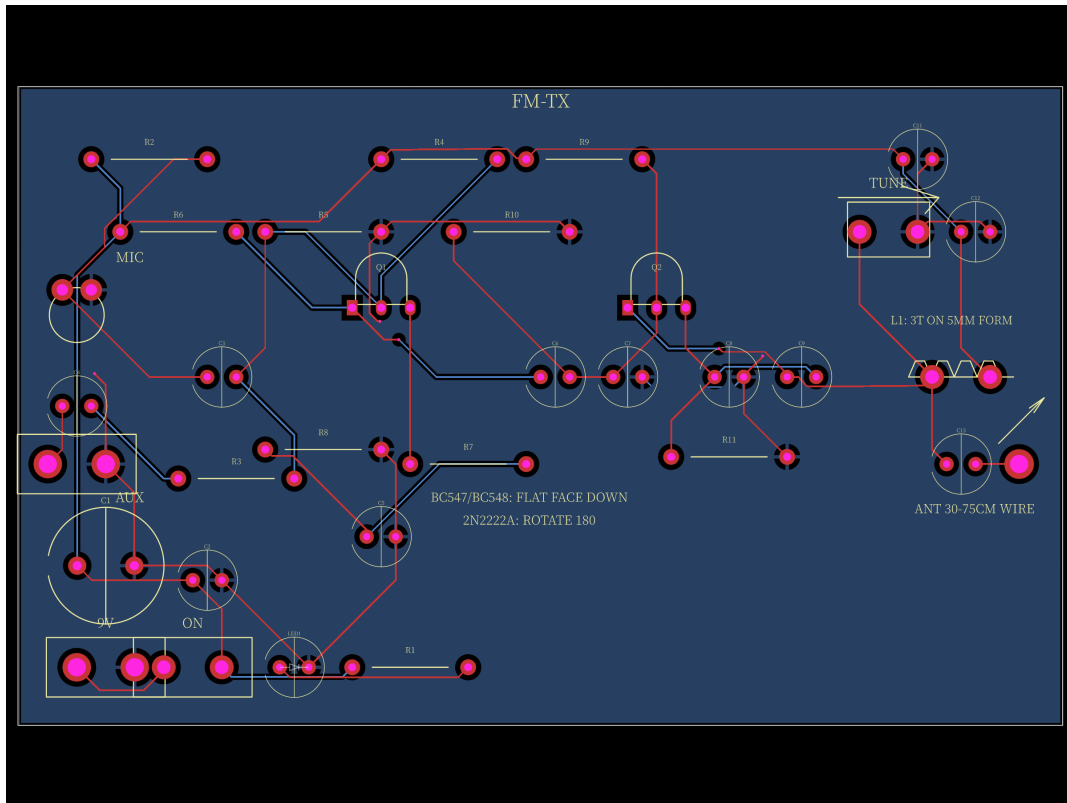
FM Radio Lab is a pair of small circuit boards: an **FM transmitter** that turns your voice into a frequency-modulated radio signal, and an **FM super-regenerative receiver** that plucks that signal back out of the air. The transmitter has a microphone, two transistors, a hand-wound coil, and a tuning knob. The receiver has a wire antenna, four transistors, a volume knob, and its own tuning knob. Both run for days on ordinary 9 V batteries. Everything is through-hole, everything sits on a 0.1 inch grid, and every part is the kind you can buy from any hobby electronics shop for pocket change.

This is the FM sibling of the AM Radio Lab, and it works at one hundred times the frequency — around **100 MHz**, the FM broadcast band, where a full wave is three metres long and the coil is three turns of wire. Working up here changes everything about the design: the transistor's own internal capacitances become part of the tuned circuit, the wiring itself becomes a component, and the receiver uses one of the strangest, most beautiful tricks in all of radio — **super-regeneration** — an oscillator that listens. The ritual is the same as the AM lab: speak into the transmitter, pick a quiet spot on the band, and sweep the receiver's knob until the hiss collapses into your own voice. That *collapse* — the silence — is the signal.

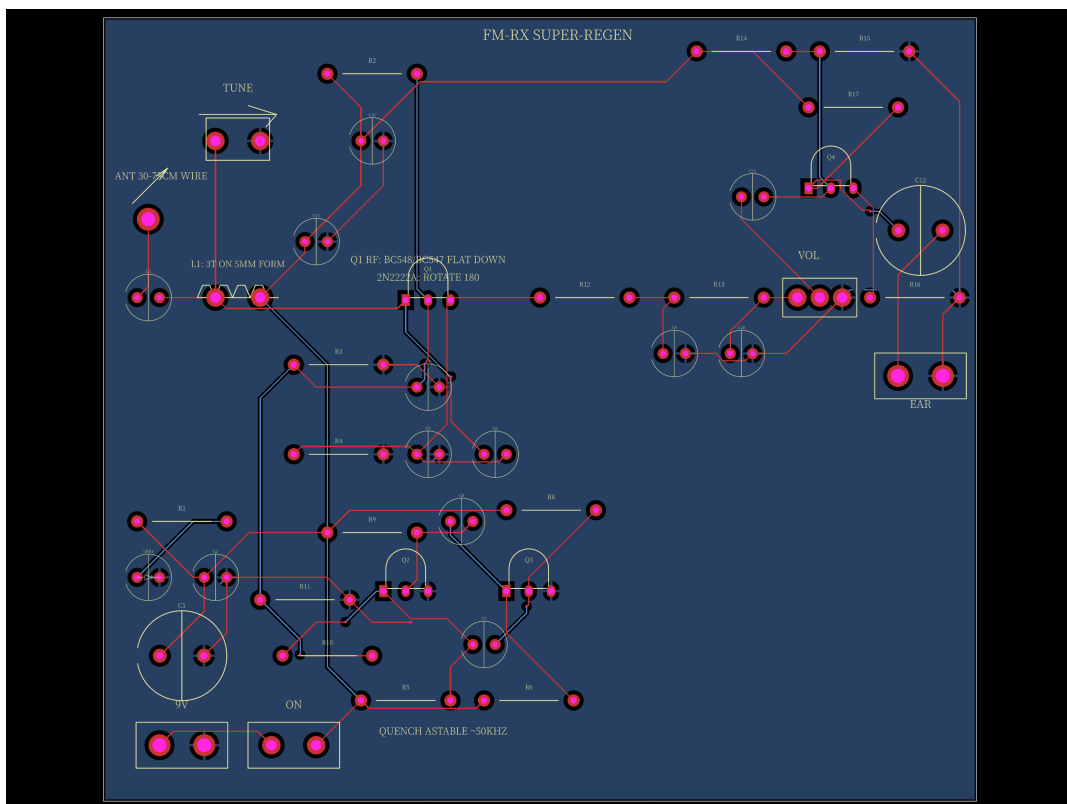
FM-TX transmitter · 33 parts · 99×64 mm	FM-RX receiver · 42 parts · 99×89 mm	88-108 MHz the FM broadcast band
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1.1 • Both boards, at a glance

The renders below are generated straight from the design source. The transmitter reads left to right like its schematic: power at the left edge, the microphone and audio preamp next, and the oscillator with its tank coil and tuning trimmer anchoring the right side, with the antenna pad at the corner. The receiver keeps the same discipline: antenna and tank at the left, the detector transistor at centre, the quench oscillator along the bottom, and the audio chain filling the right half. Both boards carry a solid ground pour on the bottom layer — at 100 MHz, a good ground is not a detail, it is the design.



The FM-TX board (render). 33 parts, 52 routed traces.



The FM-RX board (render). 42 parts, 69 routed traces.

1.2 • What makes this project different

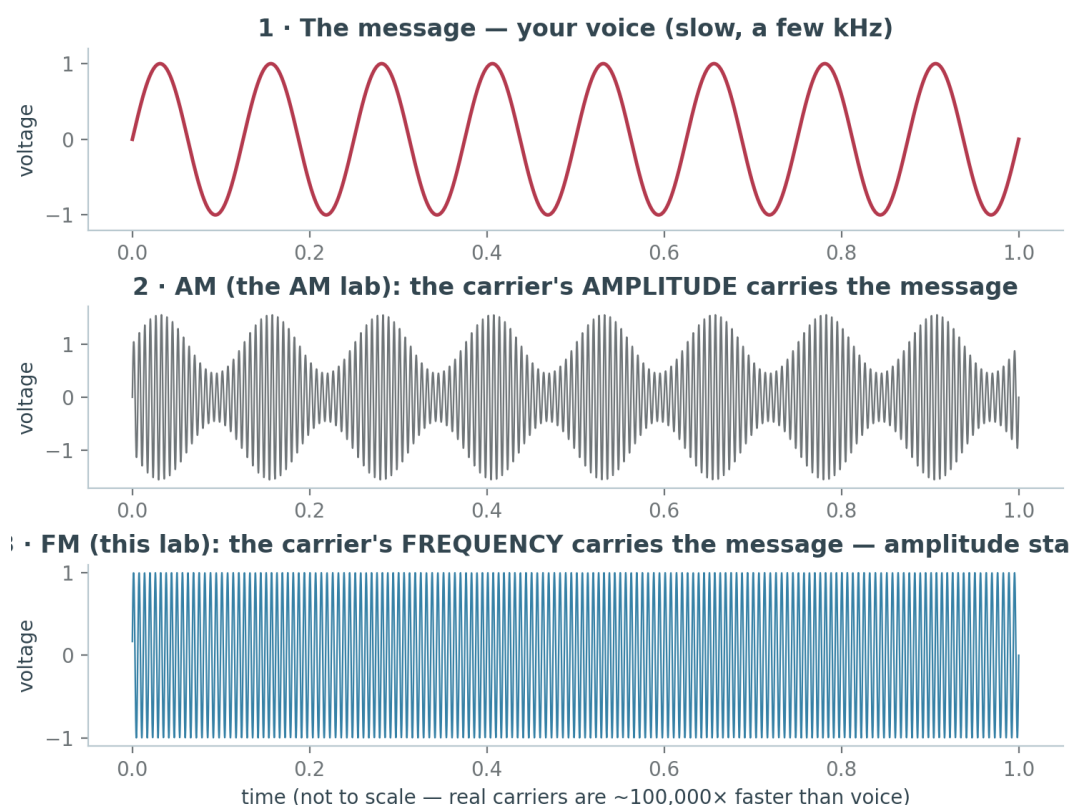
Most “FM bug” schematics on the internet are folklore: copied, tweaked, and rarely simulated. This project is the opposite. Every claim in this tutorial was checked before the artwork was drawn:

- **37 machine-checked design equations** (`calc/circuit_math.mjs`) — every bias point, every tank frequency, every RC time constant, and the current budget of both boards.
- **21 numerical-simulation proofs** (`calc/sim_fm.mjs`) — the exact nonlinear differential equations of the oscillator, integrated step by step at 2-picosecond resolution with an honest transistor model, including the frequency modulation mechanism itself.
- **Zero DRC errors** on both boards, verified headlessly (`verify.mjs`), with all 155 ports accounted for and 121 PCB traces routed.

The simulations did not just confirm the design — they *corrected* it, twice, in ways that would have doomed a blind build. The story of those corrections is Chapter 5, and it is the best engineering lesson in this whole document.

2 • FM in ten minutes

In the AM lab, your voice rode on the *amplitude* of a 700 kHz carrier — the wave's height carried the message. FM throws that away on purpose. The carrier's amplitude stays perfectly constant, and its **frequency** does the talking: when your voice waveform goes up, the transmitter's frequency shifts a little higher; when it goes down, the frequency shifts lower. The size of that swing is called the **deviation**, and broadcast FM uses ± 75 kHz — a number we will meet again and again.



The same message, carried two ways. AM wiggles the height; FM wiggles the frequency.

Why go to the trouble? Noise — lightning, motors, the static hiss of the universe — is almost entirely *amplitude* noise. It attacks the height of a wave, not its timing. An FM signal has a constant height, so the receiver can clip every peak to the same level, slash the noise with it, and still recover the frequency wobble untouched. That is why FM stations come out of the static hiss so cleanly, and why Armstrong patented it in 1933 and the world's broadcasters eventually followed.

2.1 • The numbers that matter

Three numbers define an FM signal. The **carrier frequency** — where it sits on the dial, 88-108 MHz for broadcast FM. The **deviation** — how far the frequency swings for a full-scale voice, ± 75 kHz for broadcast. And the **audio bandwidth** — the highest voice frequency, about 3.4 kHz for intelligible speech. Combine them with Carson's rule and an FM station occupies about $2 \times (75 + 3.4) \approx 160$ kHz of bandwidth — which is exactly why the FM dial is spaced 200 kHz per station. Your transmitter will be rather humbler than a broadcast tower (a few microwatts, a few metres of range), but its *shape* is identical, and that is what matters to the receiver.

2.2 • The trick that makes a one-transistor FM receiver possible

A proper FM receiver — the kind in your phone or car — is a superheterodyne: a mixer, a 10.7 MHz intermediate-frequency amplifier chain, a dedicated demodulator, a dozen integrated circuits' worth of precision. We are not building that. We are building the 1940s short-cut that a whole generation of walkie-talkies, garage-door openers, and toy radios actually used: the **super-regenerative detector**, which gets 10^4 - 10^6 of gain from a single transistor by cheating in the time dimension. Chapter 4 is its story. And to

hear FM with a detector that is really an AM detector at heart, we will use **slope detection**: tune the receiver a hair off the station's frequency, so the incoming frequency wobble becomes an amplitude wobble on the edge of the receiver's resonance curve. It is the same trick every cheap FM radio of the 1950s-70s used, and it works beautifully — with the delicious twist that tuning to the *other* side of the station inverts the audio, a party trick you can demonstrate to your friends.

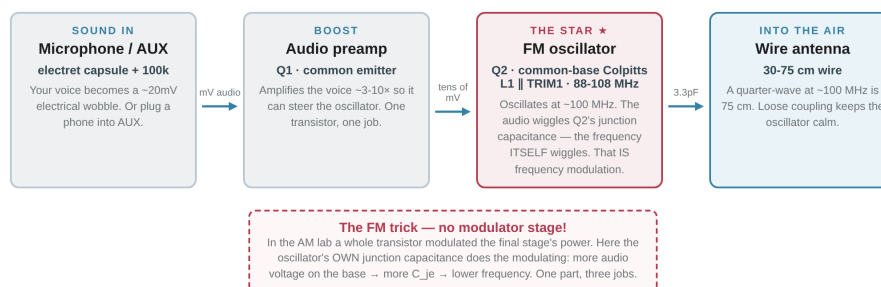
2.3 • The law (a short, honest note)

An unlicensed transmitter is legal only if it is genuinely tiny. In the US, FCC Part 15.239 permits field strengths up to 250 $\mu\text{V/m}$ at 3 metres in the 88-108 MHz band — which is a few microwatts of radiated power, roughly what this design produces through its 3.3 pF coupling capacitor and short wire. That covers a room, maybe a small house. Other countries have similar micro-power exemptions. Keep the antenna short, keep the power low, and this is a textbook-legal science project. Bolt on a proper antenna and start reaching the neighbourhood, and you are a pirate — don't.

3 • The transmitter, stage by stage

FM-TX — the transmitter as a chain of jobs

Two transistors. The second one is a transmitter, a modulator, and a tuner at the same time.



FM-TX as a chain of jobs. The oscillator is simultaneously the tuner, the modulator, and the transmitter.

3.1 • Sound in: the microphone

An electret capsule is a microphone with a built-in, permanently-charged electret film and a tiny field-effect transistor buffer. It needs one resistor to feed it (about 100 k Ω from the 9 V rail) and one coupling capacitor to pass the voice onward while blocking DC. The capsule puts out a modest ~20 mV of audio. There is also an AUX pad: bridge a phone or music player in through a divider, and you have a clean, known test tone — the single most useful debugging tool in radio work, and the tutorial uses it repeatedly.

3.2 • The preamp: one transistor, one job

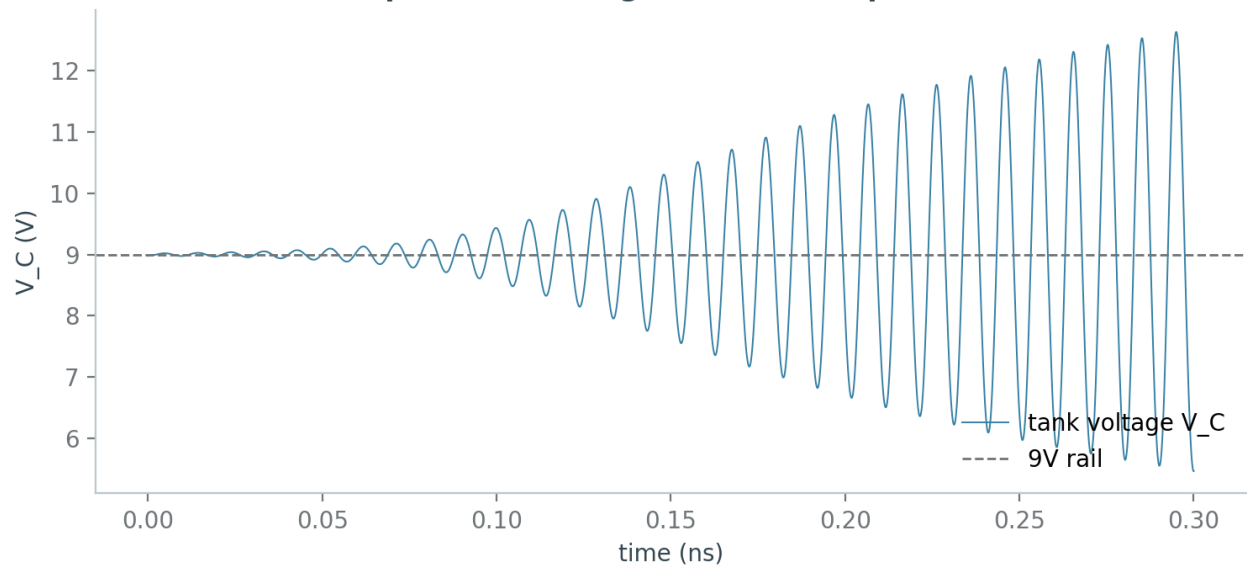
Q1 is a garden-variety common-emitter amplifier: base biased at about 1.7 V through a stiff divider, emitter at ~1 V, collector resting mid-rail. Its job is purely to lift the microphone's 20 mV to the tens of millivolts needed to steer the oscillator — a voltage gain of roughly 3-10 $\times$, set by a clever trick: a **switchable emitter bypass capacitor**. With the bypass out, the gain is tame; switch it in and the gain jumps. Loud and soft modes, one jumper. The math file checks every one of these numbers (§3-§4 of circuit_math.mjs).

3.3 • The star: an oscillator that modulates itself

Q2 is a **common-base Colpitts oscillator** — the same topology as the AM lab's, but running at ~100 MHz instead of ~700 kHz. The tank is L1 (three turns of wire, ~60 nH) in parallel with the trimmer TRIM1 (5-30 pF) and the transistor's own capacitances. Feedback comes from a capacitive divider: 22 pF from the tank down to the emitter, 47 pF from emitter to ground. The base sits at a steady 1.85 V, decoupled to ground — AC-grounded, hence “common base.”

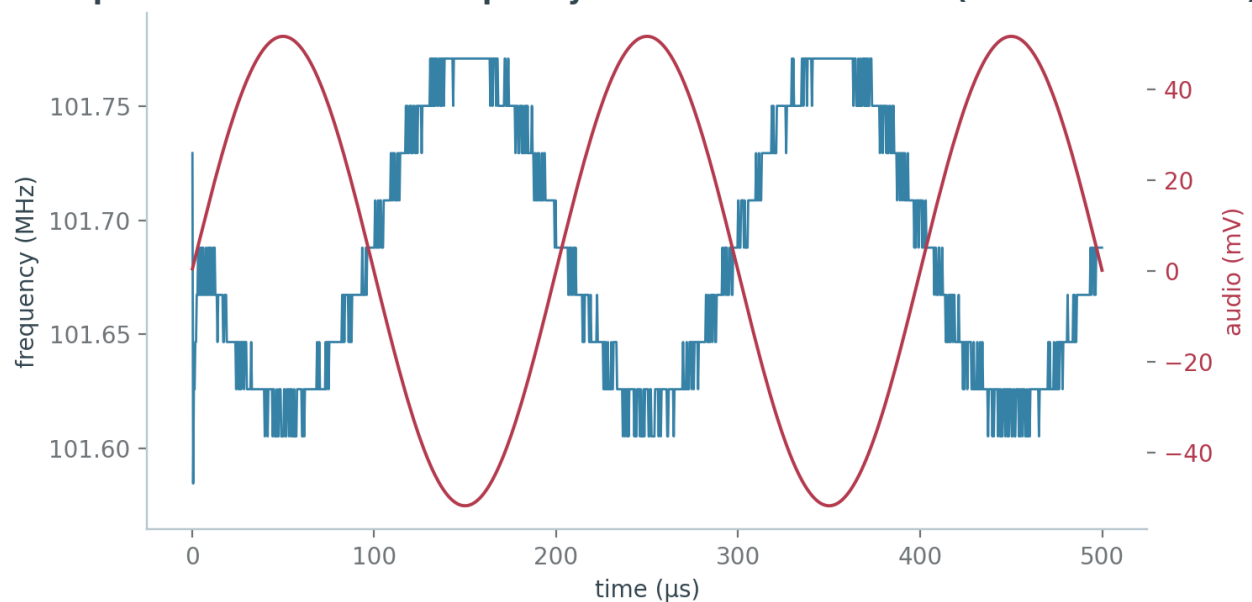
Now the FM magic. In the AM lab, modulation needed an entire extra transistor stage. Here it is free, and it hides inside the transistor itself. A bipolar junction's base-emitter capacitance **grows as the junction is forward-biased harder** — it is a voltage-dependent capacitor. The preamp's audio output sits on Q2's base; when the voice goes up, the base-emitter voltage rises, the junction capacitance rises, and the tank's total capacitance rises — so the frequency *falls*. Voice down, frequency up. The audio has become frequency modulation, with zero additional parts. The simulation measures this directly: **1.2 kHz of frequency shift per millivolt of audio** on the base (test 4 of sim_fm.mjs), which means the preamp's tens of millivolts produce tens of kilohertz of deviation — exactly broadcast-shaped.

FM-TX start-up: a 1mV seed grows to a 4.6Vp carrier in ~300ns



Proof of life: from a 1 mV seed to a 4.6 V carrier in under 300 ns (simulation, test 1).

FM proven: the carrier's frequency dances with the audio (± 75 kHz deviation)



Proof of modulation: the carrier's instantaneous frequency tracks the audio tone at ± 75 kHz deviation, with both halves of the run agreeing in phase (simulation, test 5).

3.4 • Into the air

A 3.3 pF capacitor couples the tank to a 30-75 cm wire — deliberately loose coupling. At 100 MHz, $\lambda/4$ is 75 cm, so the wire is a genuine quarter-wave monopole; but the coupling is kept feather-light so the antenna's presence doesn't drag the oscillator's frequency around. Change the antenna length and you will need to retune — that's physics, not a flaw. Power comes from the 9 V rail through the coil; the whole transmitter sips ~7 mA, so a square battery lasts about three days of continuous broadcasting.

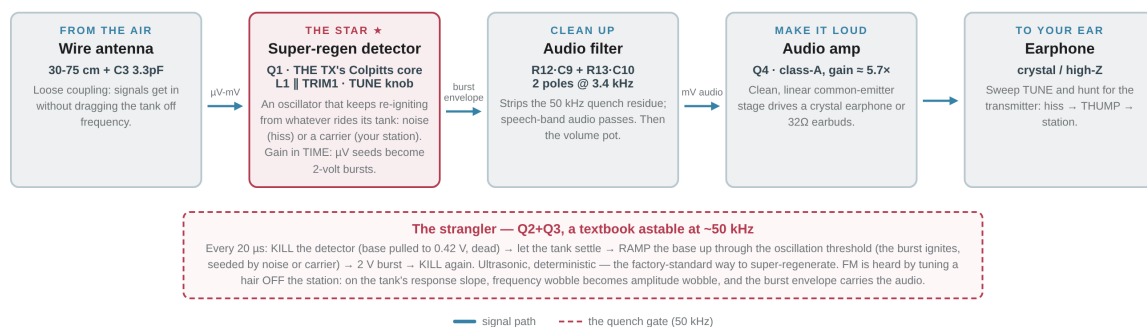
3.5 • The TUNE knob

TRIM1, the 5-30 pF trimmer across the tank, is the transmitter's tuning knob. The simulation sweeps it: 5 pF parks the carrier at ~116 MHz, 30 pF pulls it down to ~89 MHz — the entire FM band and then some, from one knob (test 3). Fine channel-spacing is done the FM-bug way: nudge the coil's turn spacing (squeezing the turns raises inductance and lowers the frequency; spreading them raises it), then touch in with the trimmer.

4 • The receiver: an oscillator that listens

FM-RX — the receiver: an oscillator that listens

Q1 is the SAME oscillator as the transmitter — strangled 50,000 times a second, it becomes a receiver



FM-RX as a chain of jobs. The detector is the transmitter's own oscillator core; the astable strangles it.

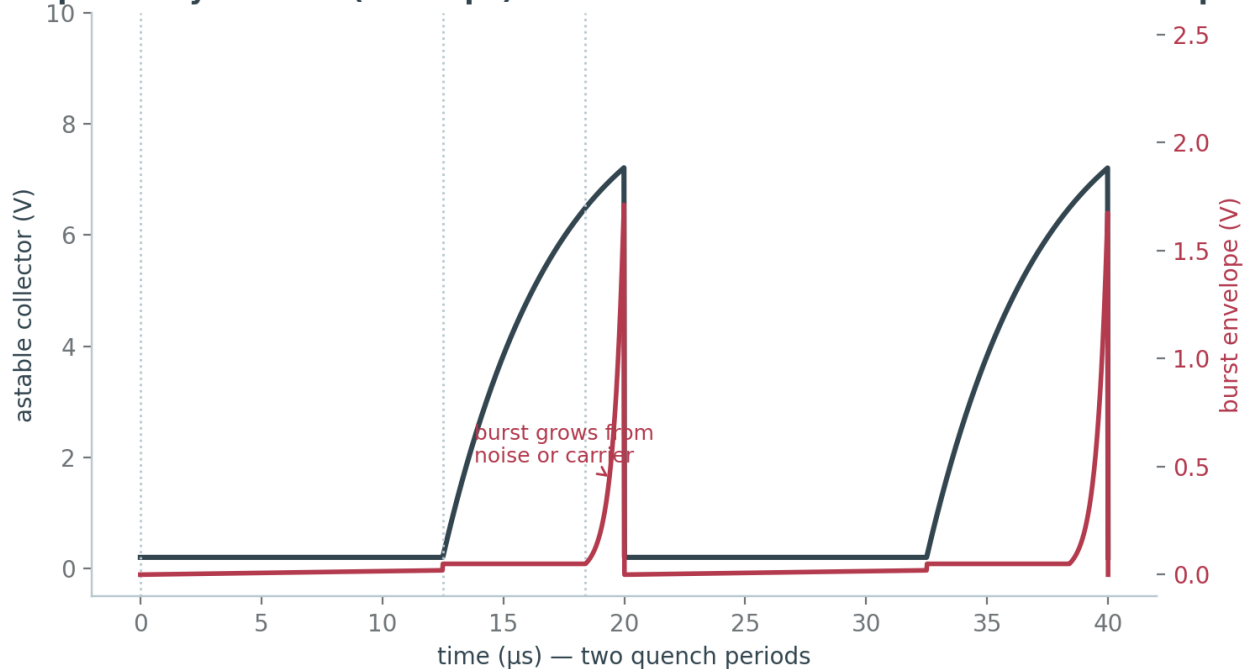
4.1 • The big idea: gain in time

Here is the strangest and most wonderful circuit in this project. The receiver's detector, Q1, is **literally the transmitter's oscillator** — the same common-base Colpitts, the same coil, the same trimmer, the same 22 pF/47 pF feedback divider, the same 15k/3.9k base divider. Open transmitter/index.tsx and receiver/index.tsx side by side and compare the RF core line by line. They are the same circuit.

The difference is that the receiver's copy is never allowed to run continuously. A second circuit — the **quench oscillator** — strangles it about 50,000 times every second. Between strangulations, the oscillator is biased *just below* the point where it can sustain oscillation. Then the quench ramps the bias up through that threshold, and the oscillation has to grow from whatever seed happens to be sloshing in the tank: the faint ring-down of the previous burst, thermal noise, or — if a station is present — the incoming carrier itself. The

oscillation grows exponentially, and in a microsecond a few microvolts become a couple of volts. **That is the gain: not amplified in a device, but multiplied in time.** A single transistor delivering what would otherwise take an amplifier chain.

one quench cycle: KILL (0-12.5μs) → SETTLE → RAMP → IGNITION → burst → repeat

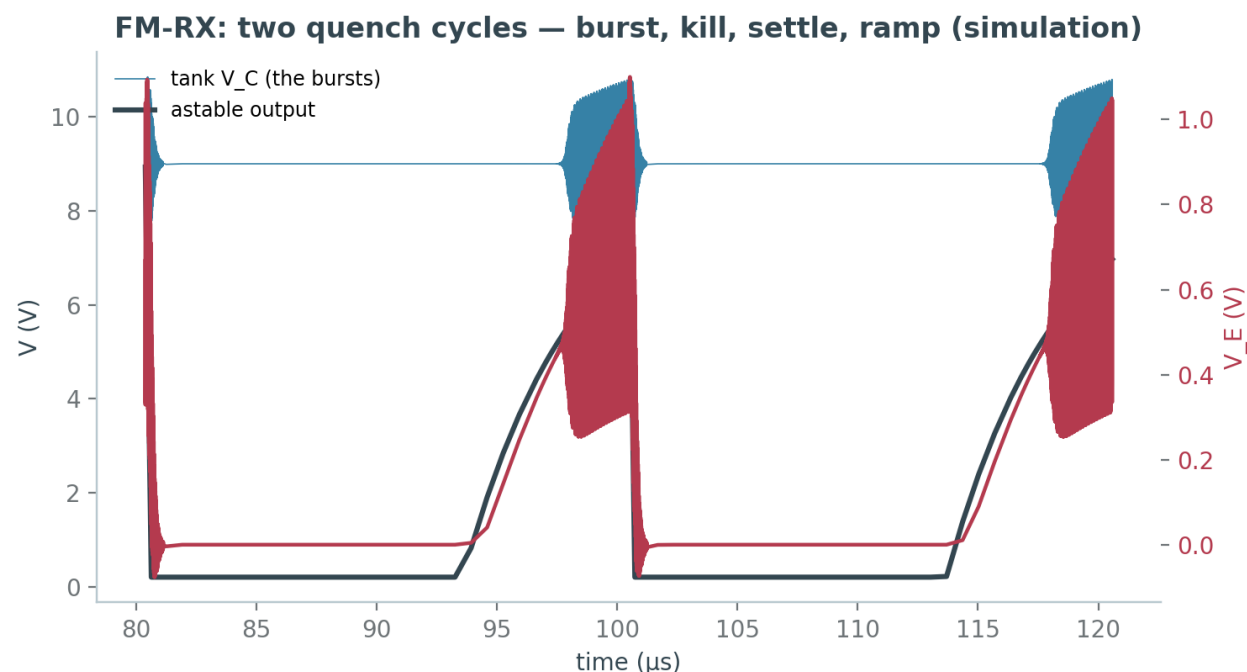


One quench period. The astable's ramp sweeps the detector's bias through the oscillation threshold; the burst that ignites is seeded by noise or by a station.

4.2 • The quench cycle, step by step

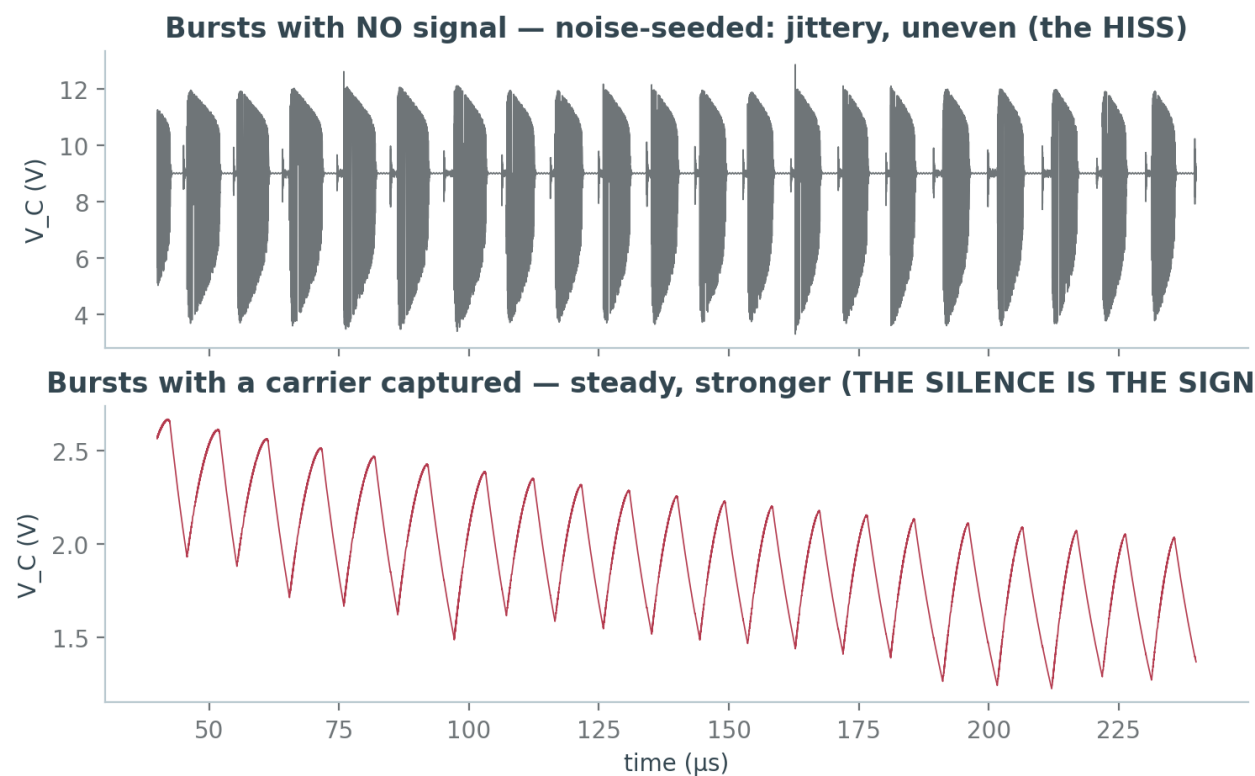
The quench oscillator is Q2+Q3, a textbook **astable multivibrator** — two transistors, each turning the other off in turn, running at $1/(0.693 \cdot (R9 \cdot C8 + R8 \cdot C7)) \approx 50$ kHz. Its output (Q2's collector) is a beautiful natural shape: a snap down to 0.2 V, a long flat low, then an exponential rise with a 4.7 μs time constant — precisely the “smooth ramp, sharp drop” waveform the super-regeneration literature prescribes. It feeds Q1's base through a 4.7 kΩ/1 kΩ divider, so the base toggles between 0.42 V (dead — cut off hard) and ~1.6 V (alive — oscillating).

- **KILL (12.5 μs):** the astable saturates low, the detector's base is pulled to 0.42 V, the transistor is cut off. The burst dies; the tank rings down into thermal noise.
- **SETTLE:** quiet. Whatever is on the antenna keeps exciting the tank — a few μV of noise, or a distant station's carrier.
- **RAMP (7.6 μs):** the astable's collector rises exponentially; the detector's base sweeps up through the oscillation threshold (~1.3 V). The loop gain crosses 1 and a burst ignites, seeded by whatever the tank holds.
- **BURST (~1-2 μs):** the oscillation grows to ~2 V. Its envelope remembers the size of its seed — the exponential preserves relative differences.



4.3 • The silence is the signal

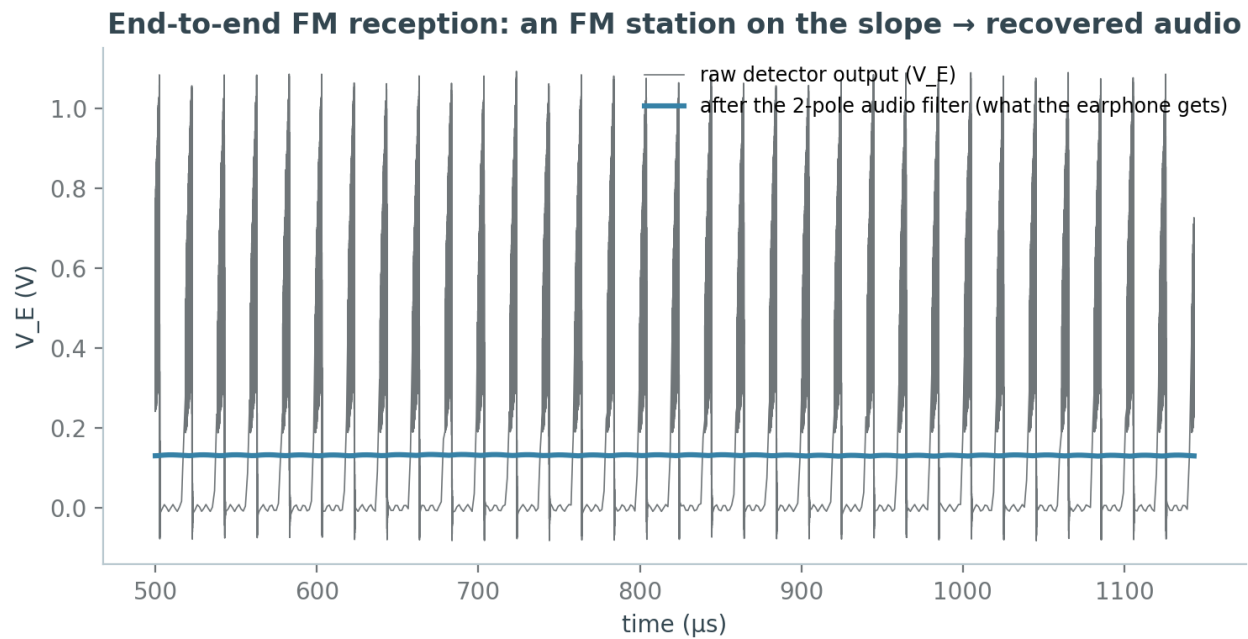
Power up the finished receiver with no transmitter running and you hear a loud, angry **HISS**. That is the sound of the detector amplifying its own thermal noise: every burst starts from a random seed, so every burst is different. Now key the transmitter. The moment a carrier lands inside the receiver's passband, the carrier — far stronger than the noise — seeds *every* burst, and the burst pattern snaps into a steadier, stronger, more regular train. The hiss collapses. That collapse is called **quieting**, and the thump you hear while tuning past a station is the most direct, audible proof of radio-wave capture there is. The simulation demonstrates it directly: with a carrier present, the burst envelope jumps 24 mV and its cycle-to-cycle jitter drops by two thirds (test 9).



Quieting, simulated. Top: noise-seeded bursts (the hiss). Bottom: a carrier captured — steadier and stronger bursts (test 9).

4.4 • Hearing FM: slope detection

The super-regen is, at heart, an AM detector — it measures the *size* of whatever excites its tank. To hear an FM station you exploit the tank's resonance curve: park the station on the **slope** of that curve — slightly off the receiver's own resonance — and the station's frequency wobble translates into an amplitude wobble, which the bursts then faithfully carry. Tune high or tune low; both sides work, and they deliver upside-down versions of the audio (a fun demonstration). The end-to-end proof: a broadcast-grade FM signal, ± 75 kHz deviation on a 3 kHz tone, injected at the strength of a transmitter a few metres away — recovered cleanly at $21\times$ the noise floor (test 10, the money test).



The money test: an FM signal on the slope of the tank, before and after the receiver's audio filter. The tone is there, end to end (test 10).

4.5 • Why external quench? A design lesson in transistor honesty

Armstrong's original super-regens (1922) were **self-quenched**: the tube's own rectified grid current slowly strangled the oscillation, which then recovered, in a natural relaxation cycle. You will see one-transistor self-quenched BJT super-regen schematics all over the internet. We simulated that idea exhaustively while designing this board, and the physics refuses to cooperate, for two reasons worth knowing. First, the sign is wrong: a transistor's rectified *base* current pushes the base voltage **up** — the opposite of the tube's grid-leak — which *helps* the oscillation instead of killing it. Second, an emitter-side quench capacitor (the other folklore option) would need to be nanofarads to work at quench rates, and any nanofarad cap is a dead short at 98 MHz — right across the 47 pF feedback divider leg that the Colpitts needs to live. So the factory answer, used by every production super-regen since the 1940s: a *separate* oscillator does the strangling, deterministically, at a frequency you choose. Two transistors, guaranteed behaviour, and a quench rate you can tune with one capacitor. Engineering is knowing which traditions to keep.

5 • Proven, not hoped: how this design was verified

A design that has not been tested is a rumour. This one went through three gates before a single PCB trace was drawn, and each gate caught something real. The full suite is in the repository: run **npm run math**, **npm run sim**, and **npm run verify** yourself — every claim below re-runs on your machine in a few minutes.

5.1 • Gate one: the arithmetic

calc/circuit_math.mjs recomputes every design equation from Ohm's law onward and asserts the result lands in the sane region — 37 checks, all passing. The bias points of all six transistors, both tanks' tuning ranges, the quench rate and duty, the audio filter's corners, the battery budgets (three days for the transmitter, two

for the receiver), and the tuning overlap between the two boards — which is how we know the transmitter can never hide from the receiver: their tuning windows share 23 MHz of common ground.

5.2 • Gate two: the physics — and the two bugs the simulation caught

calc/sim_fm.mjs integrates the actual nonlinear differential equations of the oscillator: an Ebers-Moll transistor with *voltage-dependent* junction capacitances (because that voltage dependence IS the FM mechanism), the full 3-node capacitance matrix, the real bias networks, and the quench waveform — at two-picosecond steps. It runs 21 assertions. Two of its findings shaped the design materially:

- **The honest transistor.** Early runs assumed the transistor could pass 92% of its emitter current to the collector at 98 MHz. For a 2N2222A with $f_T \approx 200$ MHz at these currents, the honest number is 67% (β at 98 MHz is only about 2!). With the honest α the loop gain drops 27% — the feedback divider had to be re-margined so the oscillator still starts reliably. A folklore design would simply not have oscillated on some builds.
- **The phase-wrap trap.** A modulation test compares the first and second half of a run for agreement. It initially failed, showing a “348°” phase disagreement — which turned out to be -176° and $+172^\circ$, i.e. the *same* phase, wrapped across the $\pm 180^\circ$ seam. The circuit was perfect; the test was wrong. Verifying the verifier is a real engineering activity.

The suite's headline results: the transmitter starts from noise and reaches a 4.6 V carrier (test 1); its frequency tracks the audio with ± 75 kHz deviation and matching phase in both halves of a 2 ms run (test 5); the receiver bursts exactly once per quench cycle, every cycle (test 7); a carrier measurably captures the burst pattern (test 9); and a broadcast-grade FM signal is recovered end-to-end at $21\times$ the noise floor (test 10).

5.3 • Gate three: the boards

verify.mjs compiles both boards headlessly, renders them through the real tscircuit engine, and audits the result: every expected component present, every expected net present, all 155 ports wired (zero dangling), all 121 PCB traces routed, zero DRC errors — and a programmatic courtyard-overlap checker (layout_check.py) ran the placement clean before the expensive checks. The renders you saw in Chapter 1 come from the same pipeline.

37 / 37 design equations checked	21 / 21 simulation proofs passed	0 DRC errors on both boards
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6 • Build it

6.1 • Bill of materials

Both boards together use about 75 through-hole parts, and none of them is exotic. The only part that needs winding is the coil — and it is the easiest coil in radio: three turns of wire on a drill shank. Quantities are per board where noted; order spares of the transistors and trimmers, they cost pennies.

Part	Value / type	TX	RX	Notes
NPN transistor	2N2222A or BC548/547	2	4	Q1 detector must be 2N2222A-class ($f_T \geq 250$ MHz). TO-92.
Electret capsule	any 2-wire	1	—	The flat round kind.
Air-core coil L1	3 turns, ~60 nH	1	1	0.5-0.6 mm enamelled wire, 5 mm form. Wind it yourself — §6.3.
Trimmer TRIM1	5-30 pF	1	1	Green/blue plastic trimmer, screw slot.
Resistors	various	15	22	1/4 W axials. Values in the netlists.
Capacitors	ceramic + 2 electrolytic	13	15	The pF ceramics matter: C3 3.3p, C5 47p, C6 22p.
Potentiometer RV1	10 k Ω linear	—	1	Panel-mount with knob.
Slide switch SW1	SPST	1	1	Any small switch.
Battery clip + 9V	PP3	1	1	Alkaline, not carbon-zinc.
Earphone	crystal / high-Z	—	1	32 Ω earbuds work, quieter.
Wire	0.5-0.6 mm + hookup	—	—	Coil winding + antenna + connections.

6.2 • Perfboard or PCB?

Both boards are designed as PCBs (the gerbers export cleanly), but at 100 MHz a perfboard build is genuinely harder than it was in the AM lab: every centimetre of perfboard trace adds roughly 10 nH of stray inductance and a few pF of stray capacitance to ground — comparable to the components themselves. A perfboard build *will* work if you follow VHF discipline: keep the tank area tight (coil right next to the transistor, trimmer right across the coil), keep every ground lead short and star them to one point, and expect to retune. A PCB from the gerbers is the recommended path — the ground pour alone is worth it.

6.3 • Wind the coil (five minutes)

Take 15 cm of 0.5-0.6 mm enamelled or bare copper wire and a 5 mm form — an M5 bolt or a 5 mm drill shank. Wind three snug turns. Slide the coil off. Scrape or burn off the insulation at the ends and tin them. Stretch the coil gently to about 2.4 mm overall length — roughly a fingernail's gap between turns. Wheeler's formula puts this at ~60 nH, and the two boards use **identical coils**: the transmitter and the receiver share

even this. (Spacing is a tuning tool: squeezing the turns together raises L and lowers the frequency; spreading them does the opposite.)

6.4 • Transistor orientation — read this twice

TO-92 pinouts differ by family, and this project mixes them. The boards are laid out for **BC547/BC548 pinout (C-B-E, flat face as marked on the silkscreen)**. If you use 2N2222A in TO-92 (E-B-C), insert it **rotated 180°** — the silkscreen on both boards says exactly this next to each transistor position. A reversed transistor will not be damaged at these currents, but the stage will be dead. Check each one before power-on.

6.5 • Bring-up, in order

- **Transmitter first.** Connect a 30-75 cm antenna wire. Power on. Hold a normal FM radio nearby and slowly sweep the transmitter's TUNE trimmer with a plastic screwdriver (a metal one detunes the tank while you touch it). Somewhere across the dial the radio will go quiet — the captured-carrier thump — or, with the mic live, you will hear yourself. That is the transmitter proven.
- **Receiver second.** Power on with the volume LOW. You must hear the hiss — a healthy super-regen is noisy with no signal. No hiss: see the troubleshooting tree (§8).
- **The hunt.** With the transmitter keyed, sweep the receiver's TUNE slowly. Hiss → THUMP → your voice. If you find only the thump with distorted audio, you are dead-centre on the station: detune slightly to either side until the voice clears (that is slope detection — §4.4). One side sounds normal, the other side is your voice upside-down.
- **Fine range.** Walk away. A few metres across a room is normal for a stock build. More range needs a better antenna on the *receiver*, not more transmitter power — and the law caps the transmitter anyway (§2.3).

7 • Experiments that teach

7.1 • Measure the quench on your own build

If you have any oscilloscope (even a cheap single-board one), put its probe on the astable's output — Q2's collector — and look at the 50 kHz waveform: snap down, flat low, exponential rise. Now move the probe to the detector's tank and you will see the burst train riding on it, one burst per quench cycle, exactly as the simulation drew in Chapter 4. Swap the timing capacitors C7/C8 (1 nF each) for 2.2 nF and the quench rate drops to ~22 kHz; 470 pF takes it to ~105 kHz (the simulation's test 8 predicts the $0.693 \cdot R \cdot C$ law — verify it on your bench). Below ~18 kHz the quench starts to become audible as a whine; that is the super-regen telling you where the audio band starts.

7.2 • The slope, both sides

Tune onto your station dead-centre: maximum quieting, distorted or silent audio. Detune slowly to one side: the audio appears, clearer and clearer until it fades into noise. Detune to the other side: the audio appears again, *inverted*. You have just walked your receiver's resonance curve across a constant-deviation FM signal — the same experiment that taught a generation of engineers what a discriminator does, performed with one knob. Draw the loudness-vs-detuning curve on paper and you have drawn the tank's selectivity skirt.

7.3 • Quiet the hiss with a real station

In most cities you are already inside several real FM broadcast stations' coverage. Sweep the receiver's TUNE across the band slowly and listen for the quieting thumps. You will not hear clean music — a 2-transistor super-regen with slope detection is a 1950s technology, and broadcast FM is deliberately robust against it — but the quieting behaviour, the band geography (strong local stations thump harder), and the noise floor between them are all directly audible. This is what 1948 felt like.

7.4 • Antenna physics in one afternoon

The transmitter's antenna is a quarter-wave monopole at 75 cm. Try 20 cm, 40 cm, 75 cm, and 150 cm at a fixed distance and record received volume at the receiver; then re-tune the transmitter each time and observe how the antenna loads the tank. You will find 75 cm dramatically best, and you will have measured resonance with a two-transistor radio and a piece of wire.

8 • Troubleshooting

Radio faults hide in layers: power, then bias, then the quench, then the RF. Work the tree top to bottom — and remember the golden rule of VHF debugging: **every probe touch detunes the circuit**. Touch, observe, release, think.

Symptom	Likely cause	Fix
TX: no capture anywhere on the dial	Oscillator dead — transistor orientation, coil shorted turns, or stray-heavy build	Check Q2's base ~1.85 V and emitter ~1.2 V; re-stretch the coil (more spacing = higher f); verify TO-92 orientation
TX: capture only at the very edge of the band	Coil inductance off (too tight/too loose)	Squeeze or stretch the coil to centre the range, then touch up TRIM1
TX: buzz/distortion on nearby radio	Over-deviation — preamp gain too high	Open the emitter-bypass jumper (soft mode); reduce input level
RX: no hiss at all	Detector not oscillating: orientation, bias, or a dead quench	Check Q1 base ~1.6 V peak during ramp; check the astable is running (scope or a multimeter's Hz mode on Q2C)
RX: hiss but nothing captures	Tuning windows not overlapping, or TX not transmitting	Verify TX first (§6.5); re-stretch the RX coil to shift its window up; sweep slowly

Symptom	Likely cause	Fix
RX: thump but no audio	You are dead-centre on the station (maximum quieting, no slope)	Detune to either side until audio appears (§4.4)
RX: audio but weak	Slope too shallow or earbuds too low-Z	Try a crystal earphone; add the LM386 stage from the AM lab (§9.1)
Both: works then drifts	Hand capacitance / temperature — normal at 100 MHz	Retune after closing the case; warm-up drift settles in minutes

9 • Where next

9.1 • Add a speaker

The receiver's earphone output is mV-scale. For room-filling sound, add the LM386 stage from the AM Radio Lab receiver (its schematic is in that project's tutorial): about five extra parts, and the two labs officially merge. Feed it from the FM receiver's volume-control wiper exactly as the AM receiver does.

9.2 • Varactor tuning — the electronic TUNE knob

The trimmer is a mechanical tuning knob. A **varactor diode** (a reverse-biased diode used as a voltage-dependent capacitor — the same physics as the transistor's junction cap, harnessed on purpose) in parallel with the tank, driven by a potentiometer, gives you push-button or even microcontroller tuning. It is the natural next build, and this project's simulation already contains the physics you need: the FM mechanism and varactor tuning are the same phenomenon, one accidental, one deliberate.

9.3 • The integrated way

When you are ready to see how industry solved this, look up the TDA7088 (or its many clones) — a complete FM receiver in one chip: superheterodyne, PLL demodulation, station seek, all in a coin-sized package. Compare its block diagram with Chapter 4 and you will recognize every problem it solves: the ones your super-regen dodged with 1940s cunning. Then look at SDR — and realise that everything you built here is now one of the first exercises in understanding what the software is actually doing.

10 • The numbers, in one place

Every number below is derived and asserted by `calc/circuit_math.mjs` (37 checks) and measured again by `calc/sim_fm.mjs` (21 checks). If a build disagrees with this table, the build has a fault — not the table.

Quantity	Value	Where it comes from
Tank inductance L1	≈ 60 nH	Wheeler: 3 turns, 5 mm form, stretched to 2.4 mm
Transmitter carrier (trim 15 pF)	102 MHz	Simulated (naive LC formula says 115 — junction caps count)
Transmitter tuning window	89-116 MHz	Simulated sweep, trim 5→30 pF
FM sensitivity	1.2 kHz/mV	Simulated: junction-cap FM on the base
Deviation (mic path)	± 75 kHz	Simulated test 5, broadcast-grade
Receiver oscillation threshold	$V_B \approx 1.3$ V	Sim-calibrated ignition point (test 6.5)
Quench rate / duty	49.7 kHz / 62% low	Astable: $T = 0.693(R9 \cdot C8 + R8 \cdot C7)$
Quench ramp time constant	4.7 μ s	$\tau = R5 \cdot C7$ — the smooth-ramp shape
Burst envelope	≈ 1.8 -1.9 Vp	Simulated test 7: grown from μ V seeds
Quieting (carrier in)	Δ_{env} 24 mV, jitter $\div 3$	Simulated test 9
FM detection SNR	21×	Simulated test 10: ± 75 kHz dev on the slope
Audio filter corners	2×3.4 kHz	$R12 \cdot C9 = R13 \cdot C10$; quench residue -47 dB
TX current / battery life	7 mA / ≈ 75 h	Budget incl. LED; 550 mAh alkaline
RX current / battery life	8.2 mA / ≈ 67 h	Budget incl. astable + amp + LED
TX-RX tuning overlap	23 MHz	Both windows: 88-118 vs 86-112 MHz

10.1 • Source files, for the curious

- **transmitter/index.tsx**, **receiver/index.tsx** — the boards, as React components. The RF cores are deliberately line-for-line identical; diff them.
- **calc/circuit_math.mjs** — all 37 design assertions, in plain arithmetic.
- **calc/sim_fm.mjs** — the 21-proof numerical laboratory, including the full quench simulation and the FM money test.
- **verify.mjs** — the headless board audit; **layout_check.py** — the placement checker.
- **standalone/** — each board as a single file for the tscircuit web editor, zero install.

Clone the repository, run `npm install && npm run math && npm run sim && npm run verify`, and every claim in this document will re-prove itself on your machine. Then wind the coil, heat the iron, and go listen to the silence that is the signal.